

ViterbiNet Revisited: A Matched-Information Classical Baseline for Learned Trellis Detection

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Abstract—ViterbiNet replaces the Viterbi algorithm’s channel-dependent branch metric with a small neural network trained from pilots and adapted online on decoded words. On Gaussian intersymbol-interference channels, its advantage was shown against Viterbi detectors given perfect, initial, or deliberately corrupted channel state information (CSI). We add a matched baseline: a Viterbi detector that uses the same pilots (one word in 25) and the same code-based acceptance rule, applied to its own decisions, and estimates its four taps by least squares. On a coded BPSK channel with COST 2100 taps, this LS-Viterbi receiver has the lowest point-estimate SER of all no-CSI receivers from 0 to 14 dB. On shared Monte Carlo draws it is significantly better than ViterbiNet with 5 adaptation updates, the best configuration here, at 1, 6, 7, and 9–12 dB and never significantly worse; 13–14 dB is not statistically resolved. It stays within about 3–30% of the perfect-CSI SER, consistent with a predicted 0.13 dB estimation loss. Under an implementable, ground-truth-free adaptation gate, LS-Viterbi keeps the lowest point-estimate SER at the three SNRs tested. The default 200 online Adam updates per word overfit; 5 updates give a lower SER at every SNR with 40 times fewer updates. A 32-parameter affine metric, the exact form of the Gaussian log-likelihood, matches the 7,002-parameter network. Only LS-Viterbi and ViterbiNet with 5 updates fit the measured detection-plus-adaptation time within a word on one CPU thread. In this experiment, the tested learned receivers show no established advantage over a classical receiver given the same information.

Index Terms—Deep learning, symbol detection, Viterbi algorithm, channel estimation, intersymbol interference, model-based deep learning.

I. INTRODUCTION

MODEL-BASED deep learning keeps the structure of a classical algorithm and learns only the parts that depend on an unknown model [1]. ViterbiNet [2] is a representative example for sequence detection over channels with memory: the Viterbi trellis [3], [4] is kept, and a small fully connected network maps each received sample to the likelihoods of the channel states. The network is trained on pilots and adapted online on decoded words that a Reed–Solomon (RS) code accepts, which lets the receiver follow a time-varying channel without new pilots. Later work added meta-learning to speed up that adaptation [5], [6]. More broadly, learned detectors have been proposed for the physical layer [7], OFDM [8], MIMO [9], and channels without tractable models [10].

On Gaussian intersymbol-interference channels, ViterbiNet was compared with the Viterbi algorithm given full CSI, which it approaches, given CSI with uncertainty, which it outperforms, and, under block fading, given only the initial CSI [2];

that work also compares learned detectors and covers non-Gaussian channels. None of these classical baselines receives the information ViterbiNet receives: they are given the true channel, a corrupted one, or a stale one, with no means of correcting it. A classical receiver that estimates the channel from the same pilots, with the same acceptance rule applied to its own decisions, is the matched comparison. Such adaptive maximum-likelihood sequence estimators are decades old [11], [12], but they are absent from the evaluations of learned trellis detectors that we are aware of.

The evaluations of ViterbiNet and of its meta-learned extensions that we are aware of [2], [5], [6] measure the learned receiver against Viterbi detectors with perfect, initial, or corrupted CSI, or against other learned receivers. This letter adds what those evaluations lack:

- *The matched-information baseline.* LS-Viterbi, a Viterbi detector given ViterbiNet’s pilots and acceptance rule, applied to its own decisions, is never significantly worse than ViterbiNet ($K = 5$) or VNet-affine in any available paired comparison, is significantly better than ViterbiNet ($K = 5$) at 1, 6, 7, and 9–12 dB, and stays close to the perfect-CSI bound, within a loss we predict in closed form (Section III-A). It keeps the lowest point-estimate SER under an implementable, ground-truth-free gate at the three SNRs tested.
- *An ablation of the adaptation budget.* The default 200 Adam updates per accepted word overfit; 5 updates give a lower SER with 40 times fewer updates (Section III-B).
- *An ablation of the metric size.* A 32-parameter affine metric, the exact Gaussian log-likelihood form, matches the 7,002-parameter network. We also measure every receiver’s cost against a real-time budget (Section III-C).

In this experiment, the tested learned receivers show no established advantage over a classical receiver that is given the same information.

II. SYSTEM MODEL AND RECEIVERS

A. Channel and frame

Information words of 120 bits are RS-encoded over $\text{GF}(2^8)$ with two parity symbols [13], BPSK-modulated ($s_t \in \{\pm 1\}$), and sent over a real intersymbol-interference channel with memory $L = 4$:

$$y_t = \sum_{k=0}^{L-1} h_k^{(w)} s_{t+L-1-k} + n_t, \quad n_t \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where $\text{SNR} = 1/\sigma^2$. The taps $h^{(w)}$ of word w are the magnitudes of a fixed COST 2100 channel trace [14] (300

blocks; within each block of 50 words the taps decay smoothly, and between blocks they jump, so every frame contains two abrupt changes); $L = 4$ matches the intersymbol-interference setting of [2]. A frame has 125 words; every 25th word is a pilot, so each frame carries 5 pilots and 120 data words. The trellis has $2^L = 16$ states $x_t = (s_t, \dots, s_{t+L-1})$.

B. Receivers

All receivers run the same 16-state Viterbi recursion and differ only in the branch metric $\ell_t(x)$ assigned to state x at time t .

Viterbi, perfect CSI: $\ell_t(x) = -(y_t - \mathbf{h}^T \mathbf{s}(x))^2/2$ with the true taps. This is the lower bound.

Viterbi, noisy CSI: the same metric with taps $1, \dots, L - 1$ perturbed by Gaussian noise whose standard deviation is a fraction ρ of the RMS tap magnitude (the CSI-uncertainty baseline of [2]).

LS-Viterbi (matched baseline, no CSI): the same metric with $\hat{\mathbf{h}} = \arg \min_{\mathbf{h}} \sum_t (y_t - \mathbf{h}^T \mathbf{s}(x_t))^2$, recomputed on every pilot word and on every data word that passes the adaptation gate, with x_t taken from that word's labels. Every adaptive receiver uses the same pilots and the same acceptance and labeling rule, applied to its own decisions.

ViterbiNet: a network $1 \rightarrow 100 \rightarrow 58 \rightarrow 16$ (sigmoid, then ReLU; 7,002 parameters) maps y_t to 16 state scores, trained with cross-entropy against the true state; the original uses hidden widths 100 and 50 [2]; 58 is this study's implementation choice. It is trained offline (25 minibatches of 300 words, Adam [15], learning rate 10^{-3}) and then takes K Adam updates on each pilot and each accepted data word, each update using the whole word. The default is $K = 200$, the number of updates used in [6] (there with mini-batches of 64); we also report $K = 5$.

VNet-affine: the network is replaced by one affine layer $y_t \mapsto \mathbf{a}y_t + \mathbf{b} \in \mathbb{R}^{16}$ (32 parameters). Up to a state-independent term, the Gaussian log-likelihood $-(y_t - \mu_x)^2/2\sigma^2$ is exactly $(\mu_x/\sigma^2)y_t - \mu_x^2/2\sigma^2$, so this is the smallest free per-state affine metric that contains the true one. (A metric parameterized by the four taps, as in LS-Viterbi, is smaller.)

The learned receivers are trained offline on words drawn from the same COST 2100 tap sequence used for evaluation, which, if anything, favors them.

Adaptation gate. We keep the gate of the code base this study builds on: a data word is used for adaptation if the SER of its RS-decoded bits is at most 0.02, and its labels are the RS re-encoded word when that SER is zero and the detector's hard decisions otherwise. Computing that SER requires the transmitted bits, so this gate is an oracle, not a receiver function; it is applied identically to every adaptive receiver. We also evaluate an implementable gate that uses no ground truth: accept a word when the re-encoded decoded word differs from the hard decisions in at most one RS symbol (bounded-distance decoding with two parity symbols), and adapt on the re-encoded word. We re-ran LS-Viterbi, ViterbiNet ($K = 5$ and 200), and VNet-affine with this gate at 7, 10, and 12 dB, from the same offline weights and on the same draws (20–30

repetitions). It accepts fewer words at 7 dB (33% vs. 44% of data words for LS-Viterbi) but labels them more accurately (15% vs. 37% of accepted words carry any label error). LS-Viterbi keeps the lowest point-estimate SER under either gate at all three SNRs. The gates are not equivalent, however: in paired differences VNet-affine is significantly worse with the RS gate at 7 dB ($+7.7 \times 10^{-4} \pm 5.6 \times 10^{-4}$); the other eleven differences are not resolved.

C. Monte Carlo protocol

Each repetition draws fresh bits and noise for one 125-word frame. The SER is computed on the RS-decoded information bits of the 120 data words (14,400 bits per repetition); the simulator's stored average also counts the five pilot words as error-free, which we remove by scaling it by 125/120. Learned receivers use at least 20 repetitions; the perfect-CSI and LS-Viterbi receivers use at least 100 (up to 10^4 and 10^3 , respectively), with the count set from a predicted SER to target about 100 errors. A point with no errors is reported as the one-sided 95% bound $3/N$ for N evaluated bits. The perfect-CSI, LS-Viterbi, ViterbiNet ($K = 5$), and VNet-affine receivers read repetition $r < 200$ from a shared cache of generated (bits, noise) draws, so their first repetitions are paired; we report paired differences over those shared repetitions with pointwise Student- t 95% intervals. The $K = 200$ ViterbiNet sweep ran with the same code on separate GPU machines with independent draws, so comparisons against it are unpaired. All reported points were generated after we corrected a data-cache defect that had replayed one draw across repetitions; results collected before the fix were discarded, and every nonzero BPSK point used here (105 points, 0–14 dB) has a repetition-to-repetition standard deviation of at least 1.9% of its mean.

III. RESULTS

A. Matched-information baseline

Fig. 1 and Table I compare the receivers from 0 to 14 dB. In point estimate LS-Viterbi has the lowest SER of all no-CSI receivers at every SNR, and it stays within about 3–30% of the perfect-CSI SER (e.g., $2.4e-2$ vs. $2.2e-2$ at 7 dB and $2.8e-7$ vs. $2.2e-7$ at 14 dB). ViterbiNet with its default adaptation is 1.3 times worse at 7 dB and 45 times worse at 14 dB. With 5 adaptation steps the gap narrows but remains: 1.07 times at 7 dB, 1.7 times at 12 dB, and 20 times at 14 dB. On the shared draws, LS-Viterbi is significantly better than ViterbiNet with $K = 5$ at 1, 6, 7, and 9–12 dB and than VNet-affine at 0, 2–7, and 11 dB (paired 95% intervals excluding zero), and never significantly worse in any available paired comparison ($K = 5$ has no complete paired record at 5 and 8 dB); at 13–14 dB the paired differences are not resolved. We report results up to 14 dB, the last SNR at which every receiver still produced errors within its simulation budget; there the ViterbiNet ($K = 5$), VNet-affine, and LS-Viterbi SERs each rest on 4 bit errors, in one or two repetitions. A Poisson interval for 4 errors (0.3–2.6 times the estimate) treats them as independent and so understates the uncertainty; the 14 dB ratios are indicative only.

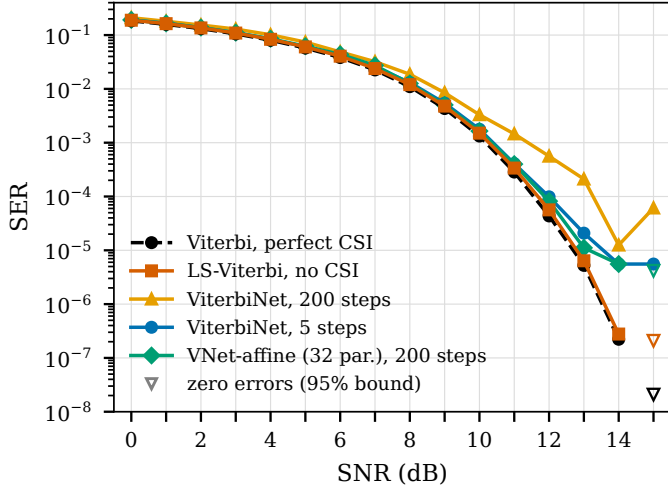


Fig. 1. SER vs. SNR. LS-Viterbi uses the same pilots and acceptance rule as the learned receivers, applied to its own decisions. Results are reported up to 14 dB; hollow triangles at 15 dB mark receivers with no errors (95% upper bound).

Why LS-Viterbi tracks the bound. With correct labels, independent noise, and taps unchanged between estimation and detection, the LS estimate from one word of $T = 136$ samples has error covariance $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, and the metric of state x sees noise of variance $\sigma^2(1 + \mathbf{s}(x)^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{s}(x))$. If the shifted-symbol columns of $\mathbf{X}$ are nearly orthogonal, $\mathbf{X}^T\mathbf{X} \approx T\mathbf{I}_L$ and the excess is L/T . We checked this for RS-coded words: averaged over 2,000 random words and the 16 states, $\mathbf{s}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{s} = 0.0301$ against $L/T = 0.0294$ (median condition number of $\mathbf{X}^T\mathbf{X}/T$: 1.39; seeded script in the code release). Treating this variance increase as an SNR shift is a heuristic, not a statement about sequence errors: to first order LS-Viterbi behaves like the perfect-CSI receiver at an SNR lower by

$$\Delta \approx 10 \log_{10}(1 + L/T) = 0.126 \text{ dB}, \quad (2)$$

or 0.129 dB with the exact average for the coded words. Shifting the perfect-CSI curve by Δ predicts SER ratios of 1.03, 1.07, 1.16, and 1.27 at 4, 7, 10, and 12 dB; we measure 1.04, 1.06, 1.13, and 1.30 (Fig. 2). The prediction ignores label errors in accepted words; that it fits this closely suggests that they have little effect and that the remaining gap is the estimation noise of four unknowns. The learned receivers instead drift away from the bound as the SNR grows, reaching 2–4 times at 12–13 dB with $K = 5$ and 13–56 times at 12–14 dB with $K = 200$: a metric fitted by a few stochastic gradient updates carries an error that is invisible at low SNR and dominant at high SNR.

The noisy-CSI baseline explains why ViterbiNet appears strong in the original comparison: with 25% tap error, the Viterbi detector improves only slowly with SNR, from 3.4×10^{-3} at 12 dB to only 1.3×10^{-3} at 14 dB (Table I), because nothing in it corrects the error. ViterbiNet corrects its model from data, and so does LS-Viterbi, with four parameters and one least-squares solve per accepted word.

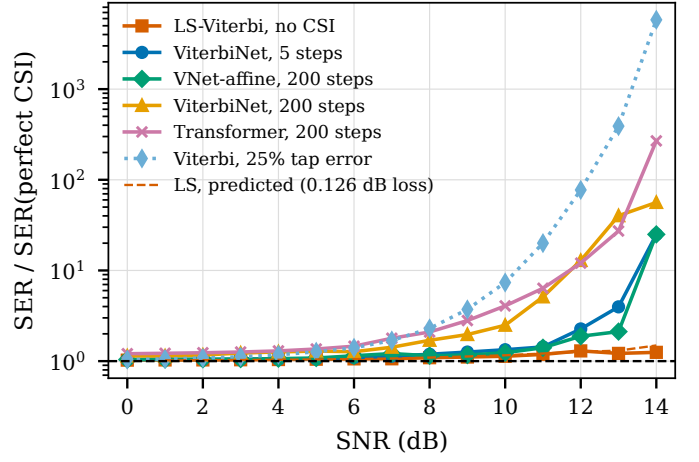


Fig. 2. SER relative to the perfect-CSI Viterbi detector. Dashed red: LS-Viterbi predicted by (2). At 14 dB the $K = 5$, VNet-affine, and LS points rest on 4 bit errors each, clustered in one or two repetitions.

TABLE I
SER OF THE RS-DECODED BITS AND ONLINE ADAPTATION COST PER ACCEPTED WORD (ONE CPU THREAD).

Receiver	Param.	7 dB	10 dB	12 dB	14 dB	Adapt.
Viterbi, perfect CSI	–	$2.2\text{e-}2$	$1.3\text{e-}3$	$4.3\text{e-}5$	$2.2\text{e-}7$	–
Viterbi, 25% CSI error	–	$3.8\text{e-}2$	$9.8\text{e-}3$	$3.4\text{e-}3$	$1.3\text{e-}3$	–
LS-Viterbi (no CSI)	4	$2.4\text{e-}2$	$1.5\text{e-}3$	$5.7\text{e-}5$	$2.8\text{e-}7$	LS solve
ViterbiNet, $K=200$	7,002	$3.2\text{e-}2$	$3.3\text{e-}3$	$5.6\text{e-}4$	$1.2\text{e-}5$	209 ms
ViterbiNet, $K=5$	7,002	$2.5\text{e-}2$	$1.8\text{e-}3$	$9.9\text{e-}5$	$5.6\text{e-}6$	5.2 ms
VNet-affine, $K=200$	32	$2.7\text{e-}2$	$1.6\text{e-}3$	$8.2\text{e-}5$	$5.6\text{e-}6$	105 ms

B. Online adaptation overfits

Fig. 3 sweeps the number of adaptation steps K at 7 dB with 25 offline minibatches. For the 7,002-parameter network the SER is lowest at $K = 5$ (0.0249) and rises steadily to 0.0315 at $K = 200$: repeated steps on a single 136-sample word fit its noise. Without adaptation ($K = 0$) the SER is 0.0340, so adaptation is useful, but a few steps suffice. From 0 to 14 dB, $K = 5$ has the lower point-estimate SER at every point (Fig. 1; these two sweeps used independent draws) and needs 40 times fewer adaptation updates per accepted word (Table I).

C. A 32-parameter metric and the latency budget

The affine metric behaves differently: it starts from a generic offline solution and needs many steps to reach the frame's channel, reaching 0.0254 at $K = 100$ –200, statistically tied with the tuned network (Fig. 3). Over the SNR sweep it matches or beats ViterbiNet with $K = 5$ from 8 to 13 dB (Fig. 2). Widening the network does not help: in a sweep of 4 to 64 hidden units and two-layer variants at 7 dB, every larger model had a higher SER than the affine one at $K = 200$ (point estimates; the smallest networks are within the confidence interval).

The parameter count matters little for latency. On one CPU thread of our implementation, the 136-step, 16-state trellis takes about 11 ms per word ($\approx 85 \mu\text{s}$ per symbol) for every receiver, while the network forward pass takes 0.05 ms

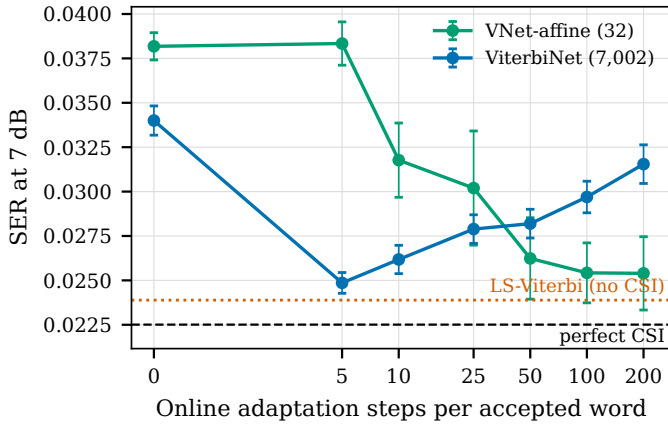


Fig. 3. SER at 7 dB vs. online adaptation steps per accepted word (25 offline minibatches; bars: 95% confidence intervals over 20 repetitions).

(affine) to 0.14 ms (7,002 parameters). The cost that differs is adaptation: one step takes 0.5 ms for the affine metric and 1.0 ms for the network, so the affine metric at $K = 200$ is not cheaper overall than the network at $K = 5$. LS-Viterbi's adaptation is a 4×4 least-squares solve (0.02 ms). At 8 ksymbol/s a word lasts 17 ms; measured detection plus adaptation of one accepted word takes 11.4 ms for LS-Viterbi and 16.8 ms for ViterbiNet with $K = 5$ (1.4% headroom), but 117 ms for VNet-affine and 220 ms for the default ViterbiNet. These are median component timings; RS decoding and gating are not included, so they do not establish a full receiver deadline. The trellis dominates the measured time only because it runs as an interpreted loop; in operation counts it costs $136 \cdot 2^{L+1} \approx 4,400$ add-compare-selects per word, less than one forward pass of the network over the word ($\approx 10^6$ multiply-accumulates), and LS re-estimation costs about 2,200.

IV. DISCUSSION AND CONCLUSION

On this channel the learned branch metric has nothing to discover: the noise is Gaussian and the channel is linear with four taps, so the correct metric is quadratic in y_t with four unknowns. A receiver that estimates those four unknowns from the same data comes within about 3–30% of the perfect-CSI SER, and no learned receiver we tested had a lower point-estimate SER. In our experiments ViterbiNet's advantage appears only against classical receivers denied that estimate. These conclusions concern our implementation of ViterbiNet and VNet-affine; we did not test the meta-learned variants [5], [6]. We therefore recommend that learned detectors be compared against a classical receiver that receives the same pilots and decisions, and that online adaptation budgets be tuned rather than fixed.

The conclusions are limited to the setting studied: BPSK, one COST 2100 tap sequence, a linear Gaussian channel, and our implementation. Learned metrics can still help where the channel is nonlinear or the noise is non-Gaussian, where a four-tap model is wrong; testing those cases against a matched-information baseline is the natural next step.

DATA AVAILABILITY AND AI-USE STATEMENT

Simulation code, raw results, and the scripts that generate every figure and table are available at <https://github.com/Gilzruk/viterbinet-revisited>. An AI coding assistant (Claude, Anthropic) was used to write simulation and plotting code and to draft the manuscript; the author checked all code, results, and text.

REFERENCES

- [1] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.
- [2] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.
- [3] G. D. Forney, Jr., "The Viterbi algorithm," *Proc. IEEE*, vol. 61, no. 3, pp. 268–278, Mar. 1973.
- [4] —, "Maximum-likelihood sequence estimation of digital sequences in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 18, no. 3, pp. 363–378, May 1972.
- [5] T. Raviv, S. Park, N. Shlezinger, O. Simeone, Y. C. Eldar, and J. Kang, "Meta-ViterbiNet: Online meta-learned Viterbi equalization for non-stationary channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2021.
- [6] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.
- [7] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.
- [8] H. Ye, G. Y. Li, and B.-H. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 114–117, Feb. 2018.
- [9] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, 2019.
- [10] N. Farsad and A. Goldsmith, "Neural network detection of data sequences in communication systems," *IEEE Trans. Signal Process.*, vol. 66, no. 21, pp. 5663–5678, Nov. 2018.
- [11] F. R. Magee and J. G. Proakis, "Adaptive maximum-likelihood sequence estimation for digital signaling in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 19, no. 1, pp. 120–124, Jan. 1973.
- [12] R. Raheli, A. Polydoros, and C.-K. Tzou, "Per-survivor processing: A general approach to MLSE in uncertain environments," *IEEE Trans. Commun.*, vol. 43, no. 2/3/4, pp. 354–364, 1995.
- [13] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *J. Soc. Ind. Appl. Math.*, vol. 8, no. 2, pp. 300–304, 1960.
- [14] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin *et al.*, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.
- [15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.